AI-Driven Waste Classification for Sustainable Waste Management

Theme: Sustainability – Waste Management

Problem Statement: Improper waste segregation is one of the biggest challenges in developing sustainable cities. Most waste is dumped without sorting, leading to pollution, increased landfill usage, and health hazards. Manual segregation is time-consuming and inefficient. Hence, there is a need for an automated, intelligent system that can classify waste into categories such as plastic, metal, paper, organic, and glass.

Goal / Objective: The goal of this project is to develop an AI-based waste classification model that helps in automated waste segregation. This supports better recycling processes, reduces pollution, and promotes environmental sustainability.

Background / Context: Sustainability focuses on maintaining ecological balance by efficiently using resources and minimizing waste. Waste classification plays a crucial role in achieving this by ensuring recyclable materials are properly separated and reused. Artificial Intelligence, particularly image classification models, can be applied to automate this process effectively.

Tools & Technology (Planned for Next Weeks):
- Python
- TensorFlow / Keras
- Google Colab
- Open-source waste classification dataset from Kaggle
- GitHub for version control

Week 1 Activities Completed:
- Attended orientation and understood project theme.
- Selected Sustainability as project theme.
- Finalized sub-theme: Sustainable Waste Management.
- Defined problem statement for AI-driven waste classification.
- Explored datasets on Kaggle for later use.
- Planned next steps for Week 2 and Week 3.

Next Steps:
- Week 2 → Collect dataset, perform preprocessing and train model.
- Week 3 → Evaluate model performance, finalize report & PPT.